# BILAL AHMED

## Senior .NET Developer | ASP.NET Core & FinTech Engineer

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## PROFESSIONAL SUMMARY

Results-driven **Senior .NET Developer & FinTech Software Engineer** with **3+ years of experience** engineering, modernizing, and scaling mission-critical brokerage back-office, ERP, and enterprise software systems. Proven expertise in building secure **RESTful & SOAP APIs**, optimizing high-volume **SQL Server databases** (stored procedures, indexing, execution plans), and integrating seamless **Angular & React** frontends. Demonstrated track record in delivering high-impact financial calculation engines (**FIFO AvgPrice**, **FIFO CGT Tax calculation**, EOD settlements, CDC/NCCPL reporting) and real-time market data feeds via **SignalR** with **99.9% uptime** on Azure cloud deployments.

## TECHNICAL SKILLS

- **Backend & Frameworks:** C#, ASP.NET Core (.NET 8/7/6), ASP.NET MVC, Web API, Web Forms, Entity Framework Core, LINQ, Dapper, SignalR, Microservices
- **Database & Performance:** MS SQL Server, PostgreSQL, MySQL, Stored Procedures, Complex T-SQL, Query Tuning, Indexing, Execution Plans, Triggers
- **Frontend & Web:** Angular, React.js, JavaScript (ES6+), TypeScript, HTML5, CSS3, Bootstrap 5, AJAX, JSON
- **Architecture & APIs:** RESTful Web APIs, SOAP Services, JWT Authentication, Clean Architecture, Repository Pattern, SOLID Principles, DI
- **Cloud, Tools & DevOps:** Microsoft Azure (App Services), Git, GitHub, IIS Server, CI/CD Pipelines, Visual Studio 2022, SSMS, Crystal Reports

## PROFESSIONAL EXPERIENCE

**Software Developer (.NET Developer)** | **Catalyst IT Solutions** — **Mar 2025 – Present | Karachi, PK**

- Architect and maintain enterprise-grade web applications using **ASP.NET Core 8, ASP.NET MVC, and C#**, ensuring **99.9% uptime** for high-availability production environments.
- Refactored and optimized **SQL Server stored procedures, indexing structures, and execution plans**, slashing query response times by **35%** on high-concurrency brokerage databases.
- Designed low-latency **real-time market watch data feeds using ASP.NET Core SignalR**, pushing sub-second live price updates to concurrent brokerage trader terminals.
- Engineered bidirectional **REST & SOAP APIs with JWT auth** for clearing houses (CDC / NCCPL) and commercial banking gateways.
- Collaborated with agile engineering teams to automate cloud deployments to **Microsoft Azure via CI/CD pipelines**.

**Associate Software Developer** | **Out2Sol IT Services & Consulting** — **Oct 2023 – Feb 2025 | Karachi, PK**

- Developed scalable line-of-business web applications using **ASP.NET MVC, C#, LINQ, and SQL Server** across diverse multi-tenant enterprise clients.
- Authored **50+ optimized stored procedures and views** to generate business-critical reports, audit trails, and analytical dashboards.
- Identified application bottlenecks, resolved database deadlocks, and refactored queries, improving transaction throughput by **25%**.
- Integrated dynamic **Angular and Bootstrap user interfaces**; configured automated **Crystal Reports** for statutory audits.

**ERP Operations Specialist (Data Systems)** | **Alkaram Textile Mills Pvt Ltd** — **Apr 2022 – Aug 2023 | Karachi, PK**

- Administered ERP data reconciliation and automated data migration pipelines from Excel to central relational databases with **99.8% record accuracy**.
- Audited inventory tracking workflows and managed chemical store modules, collaborating on relational schema designs.

## KEY ENTERPRISE & FINTECH PROJECTS

**Financial Backoffice Management System (GEM)** — ASP.NET Core & MVC | SQL Server | REST & SOAP

- Maintained and enhanced core brokerage modules: **Order Management, Risk Assessment, End-of-Day (EOD) Processing, Fund Settlements, and CDC / NCCPL Clearing**.
- Automated daily settlement, trade confirmation, and statutory audit compliance reports via **Crystal Reports**, saving 15+ hours weekly.

**FIFO AvgPrice & Capital Gain Tax (CGT) Engine** — C# | ASP.NET Core | SQL Server Stored Procedures

- Engineered automated **First-In-First-Out (FIFO)** calculation algorithm to dynamically compute real-time average buy prices across split executions.
- Implemented automated **Capital Gains Tax (CGT)** tax calculations adhering to stock exchange statutory regulations with 100% precision.

**Real-Time Market Watch System** — ASP.NET Core SignalR | WebSockets | JavaScript

- Built high-throughput real-time streaming engine pushing tick-by-tick stock quotes, volume changes, and price alerts with sub-second latency.

**Crescent Steel ERP Management System** — ASP.NET Core & MVC | C# | SQL Server | REST API

- Developed corporate **Split Share Module** automating stock split actions, instantaneously recalculating share quantities and ledger balances.

**Digital Account Opening & Insurance Portal** — ASP.NET Core | Web API | SQL Server | Angular

- Automated end-to-end customer KYC and digital onboarding workflow, **reducing account opening turnaround time by 40%**.
- Engineered policy lifecycle system covering policy issuance, automated debit/credit voucher generation, and sales commission workflows.

## EDUCATION & CREDENTIALS

**Bachelor of Science in Computer Science (BSCS)** — **Virtual University of Pakistan** — 2026 – In Progress

**Higher Diploma in Software Engineering (HDSE)** — **Aptech Computer Education** — 2021 – 2024 (Grade A)

**Intermediate in Computer Science (ICS)** — **Modern Sciences College, Karachi** — 2019 – 2021 (Grade A)

## CORE COMPETENCIES & DOMAIN STRENGTHS

• **FinTech Domain:** Stock Brokerage Operations, Order Lifecycle, Settlement Processing, CDC / NCCPL Clearing, FIFO Average Pricing, Capital Gains Tax (CGT).

• **Engineering Practices:** Clean Architecture, CQRS, Repository Pattern, Dependency Injection, Database Query Profiling, Automated CI/CD, Agile/Scrum.